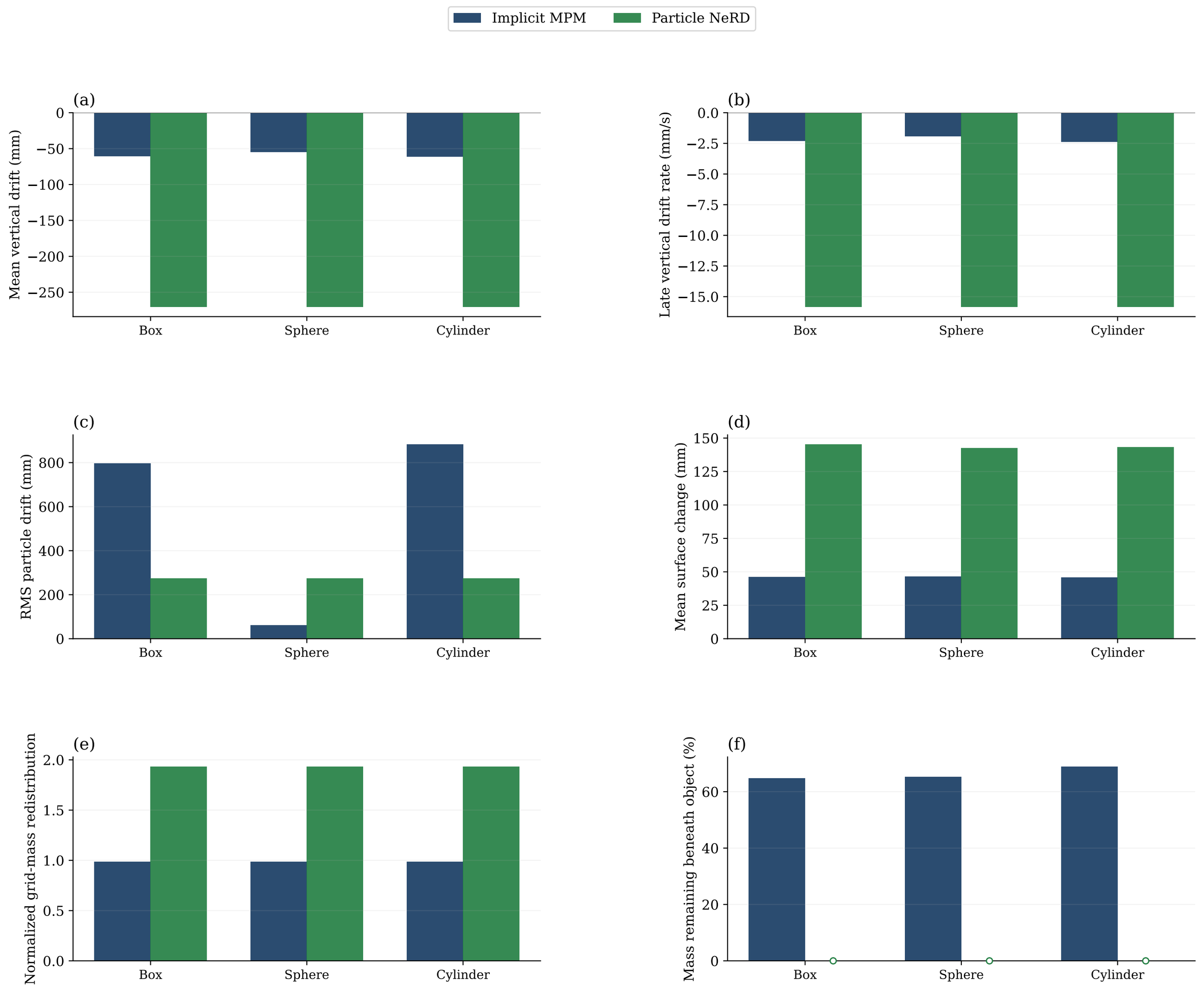

Figure 8: Frozen object-model checkpoint 9,000. One fixed-object run per shape; open circles mark measured zeros. (a) Mass-weighted mean particle displacement in world z at 23 s relative to 3 s; (b) its least-squares rate over 13–23 s; (c) RMS particle drift; (d) mean change of the material isosurface; (e) normalized absolute grid-mass redistribution; (f) mass remaining beneath the object relative to 3 s. Negative z means downward. Each solver is compared with its own 3 s state, without resetting particles. In (c), one escaped MPM particle dominates box/cylinder RMS (>99% of squared displacement); this does not describe bulk-bed drift.